THE KERNEL OF A SURJECTION ONTO A MODULE OF FINITE PRESENTATION IS OF FINITE TYPE, LOCALLY ON A RINGED SPACE

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Allowed standard prerequisites (editorial).

- The notion of a module of finite presentation: locally there is a presentation by finite free modules. The author's section on modules of finite presentation, in which the counted statement occurs. Supplies the local presentation used throughout the proof.
- The local lifting of sections of the quotient along a surjection and the generation of a kernel by displays. Used inside the counted proof at source lines 1698-1714. Produces the two maps whose combination surjects onto the kernel of the given surjection.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation.*

(1) *If $\psi : \mathcal{O}_X^{\oplus r} \to \mathcal{F}$ is a surjection, then $\mathrm{Ker}(\psi)$ is of finite type.*
(2) *If $\theta : \mathcal{G} \to \mathcal{F}$ is surjective with $\mathcal{G}$ of finite type, then $\mathrm{Ker}(\theta)$ is of finite type.*

Proof. Proof of (1). Let $x \in X$. Choose an open neighbourhood $U \subset X$ of x such that there exists a presentation

$$\mathcal{O}_U^{\oplus m} \xrightarrow{\chi} \mathcal{O}_U^{\oplus n} \xrightarrow{\varphi} \mathcal{F}|_U \to 0.$$

Let e_k be the section generating the kth factor of $\mathcal{O}_X^{\oplus r}$. For every $k = 1, \ldots, r$ we can, after shrinking U to a small neighbourhood of x, lift $\psi(e_k)$ to a section $\tilde{e}_k$ of $\mathcal{O}_U^{\oplus n}$ over U. This gives a morphism of sheaves $\alpha : \mathcal{O}_U^{\oplus r} \to \mathcal{O}_U^{\oplus n}$ such that $\varphi \circ \alpha = \psi$. Similarly, after shrinking U, we can find a morphism $\beta : \mathcal{O}_U^{\oplus n} \to \mathcal{O}_U^{\oplus r}$ such that $\psi \circ \beta = \varphi$. Then the map

$$\mathcal{O}_U^{\oplus m} \oplus \mathcal{O}_U^{\oplus r} \xrightarrow{\beta \circ \chi, 1 - \beta \circ \alpha} \mathcal{O}_U^{\oplus r}$$

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is a surjection onto the kernel of ψ.

To prove (2) we may locally choose a surjection $\eta : \mathcal{O}_X^{\oplus r} \to \mathcal{G}$. By part (1) we see $\mathrm{Ker}(\theta \circ \eta)$ is of finite type. Since $\mathrm{Ker}(\theta) = \eta(\mathrm{Ker}(\theta \circ \eta))$ we win. $\square$